

1. Administrative & Study Identification

Study Title: Sabatolimab as a Treatment for Patients With Acute Myeloid Leukemia and Presence of Measurable Residual Disease After Allogeneic Stem Cell Transplantation. **Official Title:** A Phase Ib/II, Open Label Study of Sabatolimab as a Treatment for Patients With Acute Myeloid Leukemia and Presence of Measurable Residual Disease After Allogeneic Stem Cell Transplantation **Short Title/Acronym:** STIMULUS-AML2 **Protocol Number:** CMBG453F12201 **NCT Number:** NCT04623216 **Posting ID:** 178584645724579760 **Sponsor/Company:** Novartis **Investigational Product:** Sabatolimab (monotherapy and in combination with azacitidine). Azacitidine details: ATC Code L01BC07; Trade Names: Vidaza, Onureg. **Phase of Development:** Phase 1, Phase 2 (Phase Ib/II) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To test the hypothesis that preemptive treatment with sabatolimab, alone or in combination with azacitidine, when administered to participants with Acute myeloid leukemia (AML)/secondary AML who were in complete remission with positive measurable residual disease post-allogeneic hematopoietic stem cell transplantation (MRD+ post-aHSCT), could enhance the graft versus leukemia (GvL) response and prevent or delay hematologic relapse without an unacceptable level of treatment-emergent toxicities. **Secondary Objectives:** To evaluate efficacy by measuring the incidence of aGvHD and cGvHD, and to evaluate the pharmacokinetics, particularly the Peak of Serum Concentration (Cmax) of sabatolimab.

2.2 Background & Rationale To address the high risk of hematologic relapse in patients with Acute Myeloid Leukemia (AML) or secondary AML who have achieved complete remission but remain MRD+ following allogeneic hematopoietic stem cell transplantation (aHSCT). By targeting TIM-3, sabatolimab may enhance the GvL effect and help maintain long-term remission.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Part 1 Safety Run-in) and Active Comparator/Combination (Part 2 Expansion Phase) **Blinding:** Open-label **Randomization:** Randomized (during the Part 2 Expansion Phase)

3.2 Planned Enrollment & Duration Targeted Enrollment: 24 participants (Study originally planned to enroll approximately 59 participants, but was terminated early) **Number of Sites:** Multi-center (Global) **Estimated Study Start:** 14-Sep-2021 **Trial Status:** TERMINATED (Study was terminated early)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years for adult cohorts, and ≥ 12 to < 18 years for the adolescent cohort. 2. Participants with AML/secondary AML in complete remission ($< 5\%$ bone marrow blasts, absence of circulating blasts, and absence of extramedullary disease) with MRD positivity by local or central assessment at $\geq$ Day 60 after aHSCT. 3. Systemic GvHD (graft-versus-host disease) prophylaxis or immunosuppressive treatment (IST) completely tapered for at least two weeks prior to study entry (Prednisone dose ≤ 5 mg/day or equivalent is allowed).

4.2 Exclusion Criteria 1. Any concurrent severe and/or active uncontrolled infection requiring parenteral antibacterial, antiviral, or antifungal therapy. 2. Active autoimmune disease requiring systemic therapy (replacement corticosteroids or topical/inhaled steroids are allowed). 3. Live vaccine administered within 30 days prior to the first day of study treatment. 4. Active acute GvHD grade III-IV or active moderate/severe chronic GvHD.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sabatolimab administered intravenously at dose levels of 400 mg or 800 mg every four weeks (Q4W) as monotherapy, or in combination with azacitidine for the dose expansion cohort. **Route of Administration:** Intravenous (IV) for Sabatolimab; SC/IV for Azacitidine. **Duration of Treatment:** Treatment continued until documented hematologic relapse, unacceptable toxicity, or study termination.

5.2 Concomitant & Prohibited Therapy Permitted: Topical, inhaled, nasal, and ophthalmic steroids. Replacement corticosteroid therapy (≤ 5 mg/day prednisone equivalent). **Prohibited:** Systemic immunosuppressive treatment (IST) for GvHD at study entry, and live vaccines within 30 days of Cycle 1 Day 1.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening, Informed Consent, Medical History, Fresh bone marrow aspirate for central MRD

testing | | **Baseline** | Day 0 | Randomization (if applicable),
First Dose of Sabatolimab | | **Interim** | Weeks 4-8 $\pm$ [Y] |
Dose Limiting Toxicity (DLT) observation period (≥ 8 weeks
following the first dose), Safety Labs, Efficacy Assessment | |
End of Study | Variable | Final Vital Signs, Relapse
confirmation, Overall Survival follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Rate of Dose Limiting Toxicities (Safety Run-in in Adult Sabatolimab 400mg & 800mg Only):** Assessment of tolerability of sabatolimab in adults and adolescents in the post-allogeneic stem cell transplantation setting.
- 2. Percentage of Adult Subjects With Absence of Hematologic Relapse Per Investigator Assessment (Safety Run-in and Expansion):** The percentage of adult participants for whom no evidence of hematologic relapse has been documented after 6 cycles of study treatment or earlier discontinuation at the recommended dose of sabatolimab 800 mg.
- 3. Rate of Dose Limiting Toxicities (Safety Confirmation in Adolescent Cohort Only):** Assessment of tolerability of sabatolimab in adolescent participants in the post-allogeneic stem cell transplantation setting.

7.2 Secondary Outcomes

- 1. Incidence of Grade III or IV Acute Graft Versus Host Disease (aGVHD).**
- 2. Incidence of Moderate to Severe Chronic GVHD (cGVHD).**
- 3. Peak of Serum Concentration (Cmax) Sabatolimab.**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for treatment-emergent toxicities, specifically clinically significant acute/chronic GVHD and immune-related adverse events.

Serious Adverse Events (SAEs): Standard reporting timelines (typically 24 hours) for life-threatening events or hospitalizations.

Data Monitoring Committee (DMC): Internal safety review meetings were conducted after each dose level to assess safety and determine the recommended dose for expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Set for evaluating safety, DLTs, and AEs for all patients receiving at least one dose.

Sample Size Justification: Targeted enrollment of 24 participants (originally planned ~59 participants) with a Safety Run-in to determine the safety of 400 mg and 800 mg Q4W doses in the post-aHSCT setting.

Handling of Missing Data: Standard rules for survival analysis and MRD data imputation as dictated by the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical onsite and remote monitoring to ensure safety and protocol compliance. **Audit Trail:** Data entered into validated eCRF systems, subject to source data verification and standard query resolution processes.

11. Study Identifiers & Governance

Primary ID: CMBG453F12201 **Registry Number:** NCT04623216 **Posting ID:** 178584645724579760 **Sponsor/Lead Organization:** Novartis **Study Title:** STIMULUS-AML2 **Status Description:** Study was terminated early

12. Clinical Framework (AML Specific)

Disease Area: Acute Myeloid Leukemia **Preferred Name:** Leukemia, Myelocytic, Acute **Preferred UMLS Name:** Minimal Residual Disease in Acute Myeloid Leukemia **Semantic Type:** Neoplastic Process **Diagnosis Coding:** MeSH Code: D015470 | HPO Code: HP:0004808 | SNOMED ID: 91861009 **Medical Methodology:** Pre-emptive Immunotherapy (TIM-3 Inhibition) +/- Demethylating Agent (Azacitidine) **Optimization Target:** Enhancement of Graft-versus-Leukemia (GvL) effect to prevent/delay relapse

13. Study Procedures & Methodology

Management Pathway: Post-aHSCT MRD+ treatment protocol **Education Protocol (HCP):** GvHD monitoring and TIM-3 inhibitor immune-AE training **Education Protocol (Patient):** Onsite treatment schedule & AE reporting requirements **Quality Audit Mechanism:** Central laboratory assessment for MRD conversion tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to study termination (Status: TERMINATED) **Onsite Visit Cadence:** Every 4 Weeks (Q4W) aligning with Sabatolimab infusion **Remote Monitoring Frequency:** Continuous monitoring for survival/relapse post-treatment **Checklist Review Interval:** Every cycle (28 days)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |